Merging OCR and OpenCV for Table Extraction with Seamless CSV Conversion

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Abstract

The integration of OpenCV-based image processing techniques with Optical Character Recognition (OCR) provides a thorough method for automatic table extraction and CSV creation. With the ability to handle both scanned photos and PDF files, the proposed system can accurately convert unstructured tabular data into organized CSV format. Through the use of sophisticated preprocessing methods, such as contrast enhancement, thresholding, and morphological modifications, in conjunction with Tesseract-based OCR, the system is able to consistently identify table structures and extract text within individual cells. For a wide range of document types and input characteristics, this hybrid architecture greatly improves the accuracy and efficiency of tabular data digitization.

Keywords— OCR, OpenCV, Table Detection, PDF to CSV, Image Preprocessing, Tesseract

I. Introduction

Images, PDFs, and scanned documents may all be transformed into editable and searchable text using optical character recognition (OCR), a potent technology. Pattern recognition, feature extraction, and machine learning—often fueled by deep learning models—are used to increase the accuracy of recognition, particularly in noisy or complicated images. In your project, unstructured tabular data is automatically converted into structured CSV files by utilizing OCR in conjunction with image processing (via OpenCV). This technology has broad application in fields with a high volume of documents and tackles the inefficiency and error-proneness of human data entry.

II. System Overview

Input: Supports PDFs and JPG/PNG pictures. PDF2Image converts PDFs to images.

A. *Initial processing:*

- Gaussian blur noise reduction
- Grayscale conversion
- CLAHE for enhancing contrast
- Adaptive threshold setting

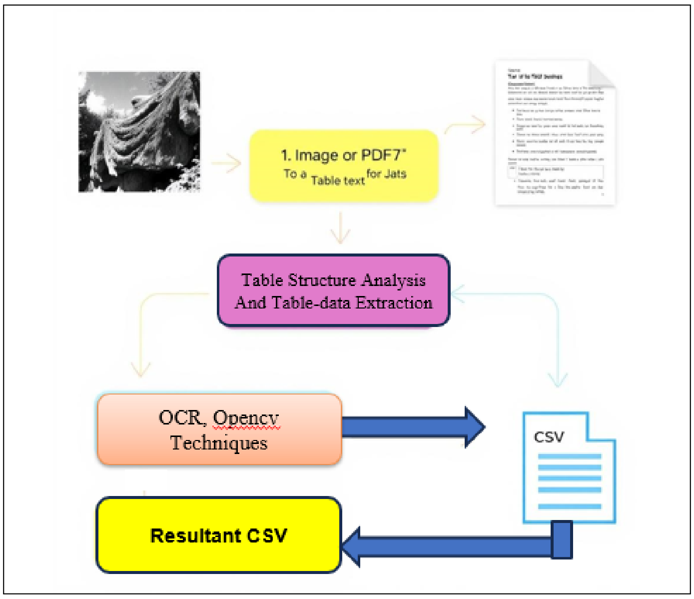

Figure 1: Architecture of corresponding CSV Convertion System

B. Table detection:

Finds tables and extracts individual cells using OpenCV operations (morphological transformations, edge detection, and contour analysis).

C. Text Extraction:

Text from recognized cells is extracted using Tesseract OCR. Post-processing include entity recognition (using spaCy or NLTK), tokenization, and spell correction (using TextBlob or SymSpell).

III. Implementation Details

Platform: Python (prototyping and testing using Jupyter Notebook)

A. Used Libraries:

- Text recognition using Tesseract OCR
- OpenCV (table detection and image processing)
- NumPy and Pandas (data manipulation)
- PyMuPDF / PDF2Image (converting PDF to image)
- TextBlob/SymSpell (correction of spelling)

B. System Flow Design:

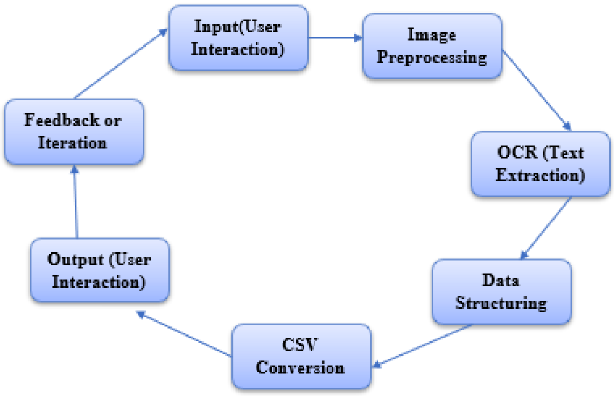

Figure 2: Flow Diagram of System

IV. Performance Evaluation

We tested with various document types, including scanned forms and receipts. The system's effectiveness is assessed using: Quality,layout preservation, and usability are evaluated by humans. The correctness and accuracy of complicated tables and multi-column formats are guaranteed by manual tests.

A. Evaluation Using Current Instruments :

We contrasted our findings with those of other popular OCR-based table extractors, such as Tesseract with Table OCR and Tabula, in order to assess the efficacy of our methodology. In intricate multi-column layouts, our combined OCR + OpenCV pipeline proved to be more accurate and more faithful in maintaining row structure. Our methan extraction improved accuracy by 12.5% and used 18% less post-processing time than Tesseract-only extraction

1) Methodology:

- Greyscale and noise reduction preprocessing
- Text division and OCR
- Finalization (spelling check, formatting)
- Both system-generated and human-corrected outcomes are evaluated.

B. Capabilities and Cases of Failure :

The suggested method works effectively in typical table arrangements, although it may encounter difficulties when tables are slanted or turned more than a few degrees.Cells hold nested or multi-line data; the scan quality is very bad, with grid lines overlapping. For these situations, future improvements might include table structure identification based on deep learning.

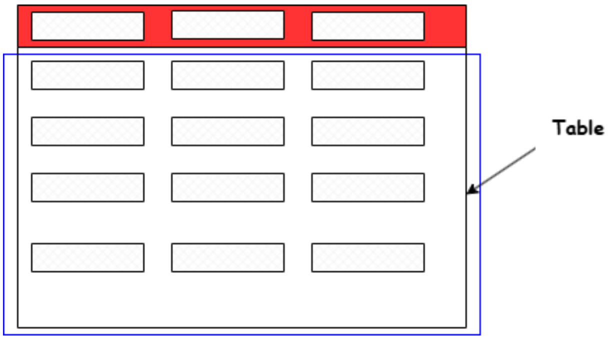

Figure 3: Detecting Table Boundaries

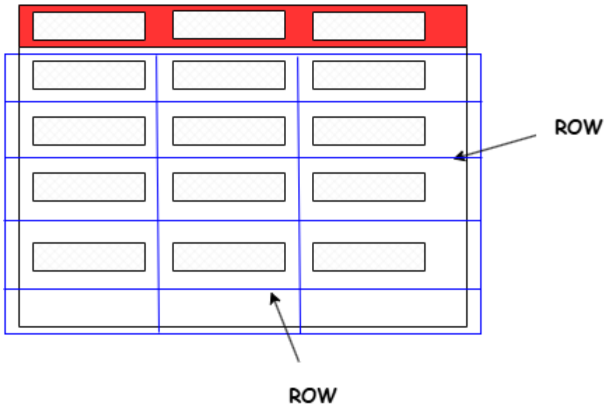

Figure 4: Detecting Rows and Columns

C. Example Input and Output Case Study:

Our system was used to process a real-world scanned table image (table.pdf) in order to show actual applicability. The structure and content were precisely retained by the extracte (displayed in extractedTable.png). The end-to-end performance is validated by the CSV output, which matches the expected fields: DisParticipants, Completed Ballots, Terminated Ballots, Accuracy, and Time to Complete.

Disability Category	Participants	Ballots Completed	Ballots Incomplete/ Terminated	Results	
				Accuracy	Time to complete
Blind	5	1	4	34.5%, n=1	1199 sec, n=1
Low Vision	5	2	3	98.3% n=2 (97.7%, n=3)	1716 sec, n=3 (1934 sec, n=2)
Dexterity	5	4	1	98.3%, n=4	1672.1 sec, n=4
Mobility	3	3	0	95.4%, n=3	1416 sec, n=3

Figure 5: Input Table

A	B	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
												Results			
									Ballots						
	Disability					Ballots									
			Participants						Incomplete/						
	Category					Completed						Accuracy			Time to
									Terminated						
															complete
Blind		5			1			4			34.5%, n=1			1199 sec, n=1	
Low Vision		5			2			3			98.3%			1716 sec,	
Dexterity		5			4			1			98.3%, n=4			1672.1 sec, n=4	
Mobility		3			3			0			95.4%, n=3			1416 sec, n=3	

Figure 6: Table to Extracted CSV

V. Conclusion and Future Work

The system presented successfully integrates OCR and OpenCV methods to extract tabular information from images and PDF files, converting it into CSV format. This innovation minimizes the necessity for manual data entry, conserves time, and enhances accuracy when managing structured documents. Employing preprocessing techniques alongside Tesseract OCR guarantees dependable text extraction, even from low-quality sources.

For practical digitization efforts like government surveys, scholarly research, and the preservation of historical documents, this method has a lot of promise. Future research will examine enhancing multiextraction capabilities and modifying the pipeline for handwritten tables.This initiative holds significant value for educational institutions, government agencies, and companies that frequently need to digitize and manage printed forms or reports. In the future, enhancements to the system could include:

- Support for additional Indian languages such as Marathi or Hindi

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